

Let N be a positive integer. For every nonnegative integer i there is a unique way to write N as

$$N = \binom{a}{i+1} + \binom{b}{i} + \dots + \binom{c}{j}$$

where $a > b > \dots > c \geq j \geq 1$. Let

$$N^{(i)} = \binom{a}{i} + \binom{b}{i-1} + \dots + \binom{c}{j-1}$$

Theorem (Kruskal-Katona): A tuple of positive integers $(1, f_0, \dots, f_d)$ is the f -vector of some simplicial complex if and only if $f_i^{(i)} \leq f_{i-1}$ for all i .

(*)

Sufficiency: The **colexicographic order** on $\binom{[n]}{i}$ is the total order defined by

$$\{a_1 < \dots < a_i\} < \{b_1 < \dots < b_i\}$$

if there is a j such that $a_j < b_j$ and $a_k = b_k$ for all $k > j$.

Given $f = (1, f_0, f_1, \dots, f_d)$ satisfying (*), let Δ_f be the set of sub consisting of the first f_i elements of $\binom{[n]}{i+1}$ for all i .

Claim: Δ_f is a simplicial complex.

Proof sketch: If

$$f_i = \binom{a}{i+1} + \binom{b}{i} + \binom{c}{i-1} + \dots$$

then the i -dim elements of Δ_f are

- all $(i+1)$ -element subsets of $\{1, \dots, a\}$
- $S \cup \{a+1\}$ where S is an i -subset of $\{1, \dots, b\}$
- $S \cup \{b+1, a+1\}$ where $\dots$ $(i-1)$ -subd of $\{1, \dots, c\}$
- $\vdots$

Define the **shadow** of some set family $A \subset \binom{[n]}{i+1}$ to be

$$\partial A := \{ S \in \binom{[n]}{i} : S \subset T \text{ for some } T \in A \}$$

If Δ_f^i is the set i -dim faces of Δ_f , then

$$|\partial \Delta_f^i| = f_i^{(i)}$$

and furthermore $\partial \Delta_f^i$ is initial in the colex order

In particular $\partial \Delta_f^i \subset \Delta_f^{i-1}$

Necessity: For sets $U, V \subseteq [n]$ with $U \cap V = \emptyset$ and $|U| = |V|$, define the (U, V) -compression

$$C_{U,V}: 2^{[n]} \rightarrow 2^{[n]}$$

by

$$C_{U,V}(S) = \begin{cases} (S \setminus V) \cup U & \text{if } V \subseteq S \text{ and } S \cap V = \emptyset \\ S & \text{otherwise} \end{cases}$$

For a family of sets $\mathcal{A} \subseteq \binom{[n]}{k}$, define $C_{U,V}(\mathcal{A})$ to be

$$C_{u,v}(A) = \{C_{u,v}(s) : s \in A\} \cup \{s : C_{u,v}(s) \in A\}$$

Then $|C_{u,v}(A)| = |A|.$

Lemma: If $A \subset \binom{[n]}{i}$ which is not initial in colex order, there exists u, v with $v > u$ in colex order such that $C_{u,v}(A) \neq A$ and

$$|\partial C_{u,v}(A)| \leq |\partial A|$$

Lemma: If you apply enough of these compressions, you will get to the initial segment in colex order.

Recall that the **geometric realization** $|\Delta|$ of a simplicial complex Δ is the topological space formed by gluing the facets of Δ along common faces.

Suppose $|\Delta|$ is some "nice" shape, like a sphere. What does that tell us about $f(\Delta)$?

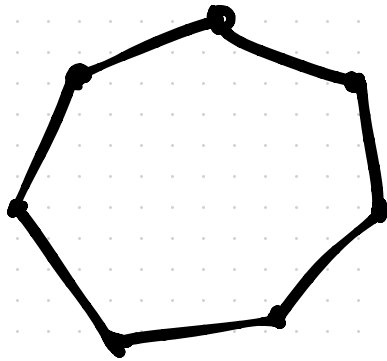
$$|\Delta| \approx S' = \text{circle}$$

$$f_0 = n$$

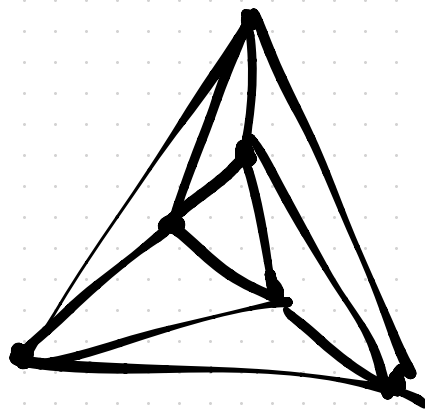
$$f_1 = n$$

All possible f -vectors are $(1, n, n)$

$$n \geq 3$$



$$|\Delta| \approx S^2$$



$$f_0 - f_1 + f_2 = 2 \quad \text{Euler characteristic}$$

$$3f_2 = 2f_1$$

$$f_0 \geq 4$$

Planar graph where
all faces are triangles

To show this is sufficient, enough
to construct Δ with each $f_0 \geq 4$

